CSE4203 Principles of Software Architecture

RapidMed

Emergency Triage and Real-Time Ambulance Dispatch System

Milestone 1: Proposal and architectural requirements

Task 6: Constraints

By Ziad Ehab

Team	RapidMed
Course	CSE4203 Principles of Software Architecture
Instructor	Dr Nada Mahmoud Hassan
Teaching assistant	Eng. Mohamed Fareh
Milestone	Milestone 1, proposal and architectural requirements
Date	23 October 2026

Team members

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6. Constraints

ID	Type	Constraint
C1	Technical	Three separately deployable parts: the paramedic app on the ambulance computer, the dispatch and routing engine, and the web portal.
C2	Technical	Three external services are fixed: the Google Routes API, Sanad’s HL7 FHIR R4 interface (a mock during the course) and Firebase Cloud Messaging.
C3	Technical	Alerts reach crews only through the ambulance computer; there is no SMS.
C4	Technical	Only synthetic patient data are used in development, tests and demonstrations.
C5	Business	Traffic-aware routing bills as Routes API Pro, with 5,000 free calls a month; a route matrix bills per element, that is, origins times destinations, and holds at most 625 elements.
C6	Business	Free service tiers only, for hosting, database and messaging; Firebase lists Cloud Messaging as no-cost.
C7	Business	Under Egypt’s Personal Data Protection Law (Law No. 151 of 2020), health data are sensitive personal data, and a breach must be reported to the Personal Data Protection Center within 72 hours.
C8	Business	Under the same law, personal data may be transferred, stored or shared outside Egypt only to countries whose protection is no lower than Egypt’s, under a licence or permit from the Center and with the data subject’s consent.
C9	Organisational	RapidMed is to become a module of Sanad, the patient health data management system of the team’s graduation project.
C10	Organisational	Three team members and four milestones: Milestone 1 is due on 23 October 2026, and Milestone 3 delivers a working prototype.
C11	Organisational	All work is submitted through GitHub; documents are PDFs named TeamName_ProjectTitle; diagram sources are kept in the repository; sources are cited in IEEE style.